

SHUBH MALHOTRA

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Education

IK Gujral Punjab Technical University

Bachelor of Technology in Computer Science & Engineering

July 2018 – June 2022

Jalandhar, Punjab

Experience

Superhero Tech Pvt. Ltd., Bengaluru

July 2025 – Present

Software Engineer, DevOps

Bengaluru, India

- Designed and operated a Kubernetes (EKS) platform serving **50+ microservices** at scale; implemented **HPA, VPA, Cluster Autoscaler**, multi-AZ managed node groups maintaining **99.95%+ uptime SLA** across all production workloads.
- Provisioned **25 production EKS clusters** via **Terraform IaC** (VPC, EKS control plane, managed node groups, IAM/IRSA, EKS add-ons); remote state in S3 + DynamoDB locking with workspaces per environment.
- Authored a shared **Helm chart library** (Deployments, HPA, PDB, Services, ExternalSecrets, CronJobs) adopted organisation-wide; reduced new microservice onboarding from **2+ days to under 30 minutes**.
- Established **GitOps delivery with ArgoCD**: app-of-apps pattern, sync waves, automated health checks and instant rollback, eliminating all manual **kubectl apply** operations in production.
- Built **Woodpecker CI** pipelines (build, test, lint, Trivy scan, ECR push) cutting PR feedback loop to **under 8 minutes** across 30+ repositories; hardened AWS security with WAF, GuardDuty, SCPs, and centralised CloudTrail for **1M+ monthly users**.
- Streamlined **Jenkins CI/CD** with SonarQube and Trivy quality gates; cut release cycle by **50%**. Deployed Prometheus (200+ targets), **Grafana RED-method dashboards**, Loki, PagerDuty alerting, driving **35% MTTR reduction**.

Superhero Tech Pvt. Ltd., Bengaluru

April 2022 – July 2025

Software Engineer, Senior SRE

Bengaluru, India

- Owned AWS infrastructure at scale: 100+ EC2 (Auto Scaling Groups), **RDS Multi-AZ PostgreSQL**, Lambda, CloudFront, EKS, ECR, handling **millions of daily API requests** against defined SLO targets.
- Architected **AWS MSK (Managed Apache Kafka)** for real-time event streaming; configured **Debezium PostgreSQL CDC connectors** for sub-second OLTP-to-analytical replication with zero data loss.
- Designed **RDS Read Replica topology** with automated failover; offloaded analytical queries reducing primary DB load by **~40%**. Engineered Jenkins multi-branch pipelines increasing **deployment frequency by 60%**, eliminating all SSH-based manual deployments.
- Led **Blue-Green deployments** with automated smoke tests and instant traffic switchback, achieving **zero-downtime releases for 18+ consecutive months**.
- Drove **20% cloud cost reduction (~\$50K/year)** via EC2 Savings Plans, Reserved Instance analysis, Auto Scaling tuning, and S3 lifecycle optimisation. Built **GitHub Actions** workflows with Trivy scanning, EKS rolling deployments and health gate verification.

Treebo Hotels, Bengaluru

Jun 2021 – April 2022

Software Engineer, SRE (Intern: Jun 2021 – Dec 2021)

Bengaluru, India

- Built production **Jenkins pipelines** integrated with AWS ECS, EC2, and S3; standardised deployment across 15+ microservices, reducing manual effort by **70%** and cutting deployment errors to near zero.
- Instrumented **Grafana dashboards** and CloudWatch alarms for SLIs (latency P50/P95/P99, error rate, saturation); improved incident detection by **30%** and drove **25% reduction in repeat incidents** through blameless post-mortems.
- Automated deployment and monitoring scripts in Python and Bash; built proficiency in PostgreSQL, AWS EC2/S3, and Linux systems, forming the foundation for subsequent SRE roles.

Key Projects

- Cost Optimization Dashboard** *Grafana · AWS CloudWatch · Cost Explorer APIs · Python / Boto3*
 - Built a multi-account cost observability platform visualising spend per service, team, and environment in real time; automated instance right-sizing via Python/CloudWatch metrics, reducing monthly cloud costs by **25%** and improving cross-team spend visibility.
- Database Migration Automation Framework** *Python · Boto3 · AWS DMS · Debezium · Kafka*
 - Engineered a fully automated Python framework for **zero-downtime database migration**: provisioned new RDS instances via Boto3, replicated schemas/tables/indexes/sequences, and configured AWS DMS tasks with full-load + CDC mode; integrated Debezium connectors for downstream Kafka consumer sync.
- Kubernetes Cluster Provisioning Automation** *Terraform · EKS · Helm · IAM/IRSA*
 - Designed reusable Terraform modules (VPC, EKS, node groups, IAM/IRSA, add-ons) to provision **25 EKS clusters** with remote S3 state, DynamoDB locking, and workspaces per environment; Helm chart library enabled new service onboarding in **under 30 minutes**.

Technical Skills

Cloud Platforms: AWS (EC2, RDS, Lambda, EKS, ECR, S3, CloudFormation, CloudFront, WAF, Cost Explorer, DMS, MSK), GCP (VPC, VM, Cloud SQL, Secret Manager)

Containers & Orchestration: Kubernetes (EKS, 25+ clusters via Terraform), Docker, Helm (custom chart authoring), HPA, VPA, Cluster Autoscaler

CI/CD & Automation: Jenkins, GitHub Actions, ArgoCD (GitOps), Woodpecker CI, Python scripting, Bash/Linux

Observability & Monitoring: Grafana, Prometheus, Loki, CloudWatch, Alert Manager, PagerDuty, Incident Management

Databases & Streaming: PostgreSQL, AWS RDS (Multi-AZ, Read Replicas), AWS MSK (Managed Kafka), Debezium CDC Connectors

AI-Assisted Engineering: ChatGPT (GPT-4o), Claude (Sonnet/Opus), Gemini (1.5 Pro / 2.0 Flash), AugmentCode, AmpCode, CodeMux; daily use for IaC generation, code review, and incident RCA

Tools & Collaboration: Jira, Confluence, Git, SonarQube, Trivy, PagerDuty, Slack, Cross-team Mentorship

Certifications & Leadership

- AWS Educate Cloud 101 · AWS Educate Solution Architect · Python Training, IIT Bombay · NASSCOM AI Practitioner
- Actively participate in hackathons and innovation sprints; deployed multiple live projects within the organisation. Hosted technical workshops as part of the ACM Chapter, Chandigarh Engineering College.
- Mentored junior engineers on Kubernetes operations, CI/CD best practices, and cloud cost optimisation strategies.